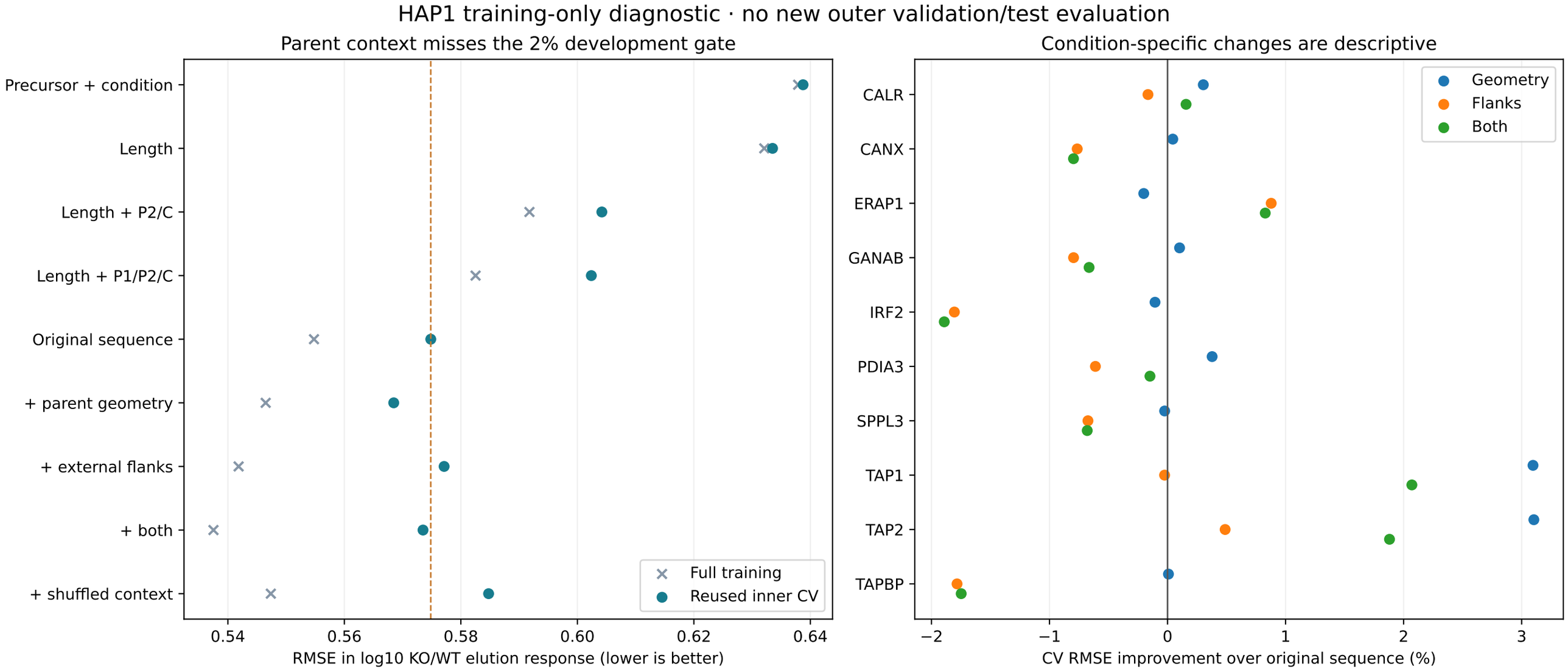
HAP1 training-only diagnostic · no new outer validation/test evaluation
Parent context misses the 2% development gate
Precursor + condition
Length
Length + P2/C
Length + P1/P2/C
Original sequence
+ parent geometry
+ external flanks
+ both
+ shuffled context
Full training
Reused inner CV
0.54
0.56
0.58
0.60
0.62
0.64
RMSE in log10 KO/WT elution response (lower is better)
Condition-specific changes are descriptive
CALR
CANX
ERAP1
GANAB
IRF2
PDIA3
SPPL3
TAP1
TAP2
TAPBP
Geometry
Flanks
Both
−2
−1
0
1
2
3
CV RMSE improvement over original sequence (%)